

=== CONFIGURACION GNB (1779966163) ===

```
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#

cu_cp:
  amf:
    addr: 10.53.1.2          # La dirección IP del AMF del Core 5G.
    port: 38412
    bind_addr: 10.53.1.10    # La IP local que el gNB usa para el tráfico del AMF (debe coincidir con la IP del
    supported_tracking_areas:
      - tac: 7
        plmn_list:
          - plmn: "00101"
            tai_slice_support_list:
              - sst: 1
    inactivity_timer: 7200    # Temporizador de inactividad para UE/Sesión PDU/DRB.
cu_up:
  ip: 10.53.1.10            # La IP del CU-UP (debe coincidir con la IP del contenedor gNB para GTP-U).

ru_sdr:
  device_driver: zmq         # Driver RF ZMQ para simulación.
  device_args: tx_port=tcp://10.53.1.20:2000,rx_port=tcp://0.0.0.0:2001,base_srate=23.04e6 # CRÍTICO ZMQ: tx al UE, rx
  srate: 23.04              # Tasa de muestreo RF para 20 MHz de ancho de banda.
  tx_gain: 75               # Ganancia de transmisión.
  rx_gain: 75               # Ganancia de recepción.

cell_cfg:
  dl_arfcn: 620000          # ARFCN para la banda n78 (3.5 GHz).
  band: 78                  # Banda NR n78 (3.5 GHz).
  channel_bandwidth_MHz: 20 # Ancho de banda de 20 MHz.
  common_scs: 15            # Espaciado de subportadoras de 15 kHz.
  ssb_scs: 15              # Espaciado de subportadoras del SSB, debe coincidir con common_scs.
  plmn: "00101"            # PLMN broadcasted por el gNB.
  tac: 7                    # Tracking Area Code (debe coincidir con la configuración del Core).
  pdccch:
    common:
      ss0_index: 0          # Índice de Search Space 0.
      coreset0_index: 12    # Índice de CORESET 0.
    dedicated:
      ss2_type: common      # Tipo de Search Space dedicado.
      dci_format_0_1_and_1_1: false # Formato DCI.
  prach:
    prach_config_index: 1    # Índice de configuración PRACH.
  pdsch:
    mcs_table: qam64         # Tabla MCS para PDSCH (64 QAM para baja latencia).
  pusch:
    mcs_table: qam64         # Tabla MCS para PUSCH (64 QAM para baja latencia).
```

```
log:
  filename: /tmp/gnb.log           # Ruta del archivo de log.
  all_level: info                 # Nivel de log para todas las capas.
  hex_max_size: 0

pcap:
  mac_enable: false               # Deshabilitar PCAPs de capa MAC.
  mac_filename: /tmp/gnb_mac.pcap
  ngap_enable: false              # Deshabilitar PCAPs de NGAP.
  ngap_filename: /tmp/gnb_ngap.pcap
  e2ap_enable: false              # Deshabilitar PCAPs de E2AP.
  e2ap_du_filename: /tmp/gnb_du_e2ap.pcap
  e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap
  e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap

metrics:
  layers:
    enable_rlc: true              # Habilitar métricas RLC.
    enable_sched: true            # Habilitar métricas del scheduler DU.
  periodicity:
    du_report_period: 1000        # Periodo de reporte de estadísticas DU (1 segundo).
```

=== CONFIGURACION UE (1779966163) ===

```
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#

[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 23.04e6 # Tasa de muestreo RF, debe coincidir con el gNB.
nof_antennas = 1

device_name = zmq # Driver RF ZMQ para simulación.
device_args = tx_port=tcp://10.53.1.10:2001,rx_port=tcp://0.0.0.0:2000,base_srate=23.04e6 # CRÍTICO ZMQ: tx al gNB, rx

[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0

[rat.nr]
bands = 78 # Banda NR n78, debe coincidir con el gNB.
nof_carriers = 1
max_nof_prb = 106 # Número máximo de PRBs para 20 MHz, 15 kHz SCS.
nof_prb = 106 # Número de PRBs para 20 MHz, 15 kHz SCS.

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 63BFA50EE6523365FF14C1F45F88737D
k = 00112233445566778899aabbccddeeff
imsi = 001010123456780 # IMSI con PLMN 00101, debe coincidir con el gNB y el Core.
imei = 353490069873319

[rrc]
release = 15
ue_category = 4
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[nas]
apn = srsapn
apn_protocol = ipv4
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[gw]
netns = ue1
ip_devname = tun_srsue
# ip_addr = 10.45.0.2          # CRÍTICO: Omitir ip_addr para que el Core asigne la IP dinámicamente.
ip_netmask = 255.255.255.0
```

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[gui]
enable = false
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=== DOCKER COMPOSE (1779966163) ===

```
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#
```

services:

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5gc:
  container_name: open5gs_5gc
  build:
    context: open5gs
    target: open5gs
    args:
      OS_VERSION: "22.04"
      OPEN5GS_VERSION: "v2.7.0"
  env_file:
    - ${OPEN_5GS_ENV_FILE:-open5gs/open5gs.env}
  privileged: true
  ports:
    - "9898:9999/tcp"
  command: 5gc -c open5gs-5gc.yml
  healthcheck:
    test: [ "CMD-SHELL", "nc -z 127.0.0.20 7777" ]
    interval: 3s
    timeout: 1s
    retries: 60
  networks:
    ran:
      ipv4_address: ${OPEN5GS_IP:-10.53.1.2} # IP fija del Core 5G.
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gnb:
  container_name: srsran_gnb
  image: srsran/gnb
  build:
    context: ..
    dockerfile: docker/Dockerfile
    args:
      OS_VERSION: "24.04"
  privileged: true
  cap_add:
    - SYS_NICE
    - CAP_SYS_PTRACE
  volumes:
    - /dev/bus/usb/:/dev/bus/usb/
    - /usr/share/uhd/images:/usr/share/uhd/images
    - gnb-storage:/tmp
  configs:
    - gnb_config.yml
    - gnb_compose_config.yml
  networks:
```

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    ran:
      ipv4_address: ${GNB_IP:-10.53.1.10} # IP asignada al contenedor gNB.
    metrics:
      ipv4_address: 172.19.1.3
    depends_on:
      5gc:
        condition: service_healthy
    command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

ue:
  container_name: srsran_ue
  image: srsran/ue # Asumiendo una imagen srsran/ue disponible.
  build:
    context: ..
    dockerfile: docker/Dockerfile # Usar el mismo Dockerfile base y adaptar el comando.
    args:
      OS_VERSION: "24.04"
  privileged: true
  cap_add:
    - SYS_NICE
    - CAP_SYS_PTRACE
  volumes:
    - /dev/bus/usb/:/dev/bus/usb/
    - /usr/share/uhd/images:/usr/share/uhd/images
    - ue-storage:/tmp
  configs:
    - ue_config.conf
  networks:
    ran:
      ipv4_address: ${UE_IP:-10.53.1.20} # IP asignada al contenedor UE.
    depends_on:
      gnb:
        condition: service_started # El UE necesita que el gNB esté iniciado.
    command: ue -c /ue_config.conf

configs:
  gnb_config.yml:
    file: ./configs/gnb_config.yml # Ruta al archivo de configuración del gNB.
  gnb_compose_config.yml:
    content: |
      cu_cp:
        amf:
          addr: ${OPEN5GS_IP:-10.53.1.2}
          bind_addr: ${GNB_IP:-10.53.1.10} # Debe coincidir con la IP del contenedor gNB.
    metrics:
      autostart_stdout_metrics: true
      enable_json: true
    remote_control:
      bind_addr: 0.0.0.0
      enabled: true
  ue_config.conf:
    file: ./configs/ue_config.conf # Ruta al archivo de configuración del UE.

volumes:
  gnb-storage:
  ue-storage:

networks:
  ran:
    ipam:
      driver: default
      config:
        - subnet: 10.53.1.0/24 # Subred obligatoria para la red RAN.
  metrics:
    ipam:
      driver: default
      config:
        - subnet: 172.19.1.0/24

```